Faculty Mobile App UI & Feature Specification

Class Attendance Management System (am-oss) • Mobile UI Architecture Guide

Target Audience:	UI/UX Designers & Mobile Developers	**Date:**	September 2026
System Core:	am-oss / Flutter Mobile / FastAPI	**Visual Mode:**	Dark Slate & Royal Blue (High Contrast)

1. Executive Summary & Core Business Rules

The Faculty Mobile Application is the authoritative academic attendance portal designed for professors and lecturers. It supports both live biometric facial recognition during scheduled classes and an attendance entry window for completed classes. Strict temporal and role boundaries govern access to periods:

Period Window	Business Rule / Temporal Condition	Visual Status	Action Allowed
Today's Completed	Markable strictly after scheduled end time (currentTime >= endTime)	**AVAILABLE**	**Mark Attendance** (Green button)
Today's Active	Ongoing period right now (startTime <= currentTime < endTime)	**LOCKED**	Locked (Displays unlock time)
Today's Future	Scheduled for later today (currentTime < startTime)	**LOCKED**	Locked (Displays unlock time)
Yesterday's Completed	All scheduled periods from yesterday (24-hour grace window)	**AVAILABLE**	**Mark Attendance** (Green button)
Already Recorded	Attendance already submitted previously by authorized faculty	**RECORDED**	**Attendance Recorded** (Outline review)
Dates < Yesterday	2+ days in the past (Tamper-prevention protocol)	BLOCKED	Hidden from Faculty app
Future Dates	Tomorrow and beyond	BLOCKED	Hidden from Faculty app

2. Design Tokens & Visual Hierarchy

The interface adopts a high-contrast dark palette tailored for mobile OLED displays and campus environments. Standardized color tokens ensure instant status recognition across all screens:

Token	HEX Value	Role & Semantic Meaning	UI Placement Examples
Primary Brand	#2563EB	Royal Blue: Brand identity, primary CTAs	Submit button, Active tabs, Profile card
Dark Canvas	#0F172A	Dark Slate: Main scaffold background	Global screen background
Card Surface	#1E293B	Slate Surface: Container cards & sheets	Class cards, Header cards, Modals
Elevated Surface	#334155	Lighter Slate: Dividers, borders, inputs	Search input, unselected buttons, borders
Status OK	#10B981	Emerald: Present, Available, Success	Available pill, [P] chip, Open button
Status Critical	#EF4444	Red: Absent, Error, Sign Out	[A] chip, All Absent button, Sign Out
Status OD	#8B5CF6	Violet: On-Duty (Official Excusal)	[OD] chip, OD reason badge, OD dialog
Status Warning	#F59E0B	Amber: Locked periods, Alert banner	Locked pill, Password change alert

Text Primary	#F8FAFC	Pure White: High-contrast headings	Headers, student names, button text
Text Muted	#94A3B8	Slate Grey: Secondary captions	Roll numbers, timing labels, subheaders

3. Global Navigation Structure

The app utilizes a **Persistent Scaffold Architecture** comprising a top AppBar, scrollable body area, and a 4-tab Bottom Navigation Bar:

- **Tab 0 - Dashboard:** Overview hub, active class card, attendance shortcut, today's schedule.
- **Tab 1 - Attendance:** Dual-tab Available Periods window (TODAY & YESTERDAY).
- **Tab 2 - Timetable:** 7-day horizontal weekday bar with complete period schedule.
- **Tab 3 - Profile:** Faculty identification, credentials, server status, and secure sign out.

4. Detailed Screen Specifications & Button Locations

Screen 1: Faculty Authentication & Login

Layout & Elements:

- **Top Right Server Pill:** Pill badge indicating backend connection status. Tapping opens *Server Configuration Dialog*.
- **Role Segmented Tab:** Full-width toggle between STUDENT and FACULTY. Faculty segment activates blue highlight (#2563EB).
- **Identifier Input & Live Validation Pill:** Text input with real-time 500ms debouncing. When valid, displays green check badge '✓ Faculty account recognized' directly below the field.
- **Password Input & Unmask Button:** Secure text field with trailing eye icon (■) button to reveal/hide password characters.
- **'Faculty Sign In' Button:** Centered at bottom of form, full width (48dp height, #2563EB). Shows centered progress spinner during authentication.

Screen 2: Faculty Dashboard Screen (Main Hub)

Layout & Elements:

- **AppBar Actions:** Title 'Faculty Dashboard' centered. Top right Logout icon button triggers sign out confirmation.
- **Faculty Profile Card:** Unconditionally rendered at top in deep navy (#2563EB). Left avatar circle with initial ('S'), faculty full name ('Dr. Shankar'), and ID subtitle ('Faculty ID: fac04').
- **Mandatory Password Banner (Conditional):** Amber alert (#FEF3C7) with right-aligned '[Update]' button.
- **Active Scheduled Slot Card:** Central prominent card. Top right displays 'ACTIVE NOW' (green dot) or 'NO ACTIVE CLASS' (grey). Shows Class, Period, Subject Code, Time interval.
- **'START ATTENDANCE SESSION' Button:** Full-width elevated button inside active slot card (height 44dp, #2563EB). Starts live session & opens Live Session Screen.
- **Mark Attendance Quick Access Card:** Checklist icon with green right-aligned '[Open]' button (#10B981). Directly switches bottom nav to Tab 1 (Attendance Window).

• **Today's Schedule Section:** Section header with right-aligned '[View Full Week]' link (switches to Tab 2). Lists period cards or clean empty state.

• **Error Recovery Card:** In case of network glitch, displays cloud-off icon, error explanation, and an active '[Retry]' button calling loadDashboardData().

Screen 3: Faculty Attendance Window (Available Periods)

Layout & Elements:

• **Sub-Tab Bar:** DefaultTabController with 'TODAY' and 'YESTERDAY' tabs pinned below AppBar.

• **Notice Banner:** Informational card explaining that completed periods unlock after end time, and yesterday's periods are open for attendance.

• **AVAILABLE Period Card:** Green accent border. Period number badge ('P1'), time ('09:10 - 10:00'), status pill ('AVAILABLE'). Features full-width green '[MARK ATTENDANCE]' button (#10B981) which opens Attendance Entry Screen.

• **LOCKED Period Card:** Amber status pill ('LOCKED'). Action button is replaced by an informational lock banner with clock icon: 'Attendance opens after this period ends at 15:20'.

• **RECORDED Period Card:** Royal blue status pill ('RECORDED'). Features full-width green outline button '[ATTENDANCE RECORDED]' allowing faculty to review or correct records.

Screen 4: Faculty Attendance Entry Screen (Roster Marking)

Layout & Elements:

• **Header Summary Card:** Pinned at top. Displays period badge, time, date, subject code/name, and class section.

• **Live Stat Counters:** 4 responsive counters: Total (White), Present (Green), Absent (Red), OD (Violet). Update in real-time as students are marked.

• **Batch Action Buttons:** Right-aligned '[All Present]' (green outline) and '[All Absent]' (red outline) buttons for rapid batch roster marking.

• **Search Input Field:** Real-time filter by student name or roll number.

• **Student Roster Row:** Student index (#1), full name, roll number, and segmented 3-button group:

  - **[P] (Present):** 36x36dp square button. Fills solid emerald green (#10B981) when selected.
  - **[A] (Absent):** 36x36dp square button. Fills solid coral red (#EF4444) when selected.
  - **[OD] (On-Duty):** 36x36dp square button. Fills solid violet (#8B5CF6) and opens OD Reason Dialog.

• **OD Reason Tag:** Displays under student name with edit icon (✏) showing selected reason (e.g. 'OD: Sports Event').

• **Sticky Bottom Submit Bar:** Pinned to screen bottom (height 48dp, #2563EB). Full-width '[Submit Attendance]' button submits payload and refreshes attendance window.

Screen 5: Live Session Screen (Biometric & Manual Attendance)

• Displays live session ID, subject, and class. Features '[TAKE GROUP PHOTO]' (camera) and '[UPLOAD PHOTO]' (gallery). Face recognition engine highlights recognized students; faculty can confirm candidates or manually toggle attendance before tapping '[Finish Session]'.

Screen 6: Faculty Timetable Screen

• Horizontal 7-day selector (Mon-Sun). Current day has active dot indicator. Lists all assigned teaching periods with period badge, times, and subject metadata. Includes pull-to-refresh and retry error card.

Screen 7: Faculty Profile Screen

• Displays large avatar, full name, employee ID, active status pill, registered email, and connected server endpoint. Includes '[Change Password]' modal trigger and prominent red '[Sign Out]' button with confirmation dialog.

5. Master Interactive Component Placement Map

The following table catalogs every interactive button, exact screen location, visual styling, and triggered action:

Button Label	Screen / Section	Visual Style & Color	Tap Action & Destination
Configure Server	Login Screen (Top Right)	Icon button, #94A3B8	Opens Server Settings Dialog
Role Segment	Login Screen (Upper)	Tab selector, #2563EB	Switches between Student/Faculty login
Unmask Password	Login Screen (Input)	Trailing eye icon (■)	Toggles password character visibility
Faculty Sign In	Login Screen (Bottom)	Elevated 48dp, #2563EB	Validates credentials & opens Dashboard
Logout	AppBar (All screens)	Icon button, White	Displays Sign Out Confirmation Dialog
Resume Session	Dashboard (Banner)	Elevated, #4F46E5	Navigates to live FacultySessionScreen
Start Session	Dashboard (Active Slot)	Elevated 44dp, #2563EB	Initiates live class attendance session
Open Attendance	Dashboard (Quick Access)	Elevated 36dp, #10B981	Switches Bottom Navigation to Tab 1
View Full Week	Dashboard (Schedule)	Text button, #2563EB	Switches Bottom Navigation to Tab 2
Retry	Dashboard / Timetable	Elevated with icon, #2563EB	Retries failed API network fetch
Sub-Tabs (Today/Yest)	Attendance Window	DefaultTabController, #2563EB	Toggles Today vs Yesterday period list
MARK ATTENDANCE	Attendance Card (Available)	Elevated 44dp, #10B981	Opens Faculty Attendance Entry Screen
ATTENDANCE RECORDED	Attendance Card (Marked)	Outlined 44dp, #10B981	Opens Entry Screen in review/edit mode
All Present	Entry Screen (Action Bar)	Outlined with icon, #10B981	Sets all students in roster to PRESENT
All Absent	Entry Screen (Action Bar)	Outlined with icon, #EF4444	Sets all students in roster to ABSENT
[P] Chip	Entry Screen (Student Row)	Square 36x36dp, #10B981	Toggles student attendance to PRESENT
[A] Chip	Entry Screen (Student Row)	Square 36x36dp, #EF4444	Toggles student attendance to ABSENT
[OD] Chip	Entry Screen (Student Row)	Square 36x36dp, #8B5CF6	Opens On-Duty (OD) Reason Dialog
Confirm OD	OD Reason Dialog	Elevated, #8B5CF6	Saves OD reason & marks student OD
Submit Attendance	Entry Screen (Sticky Bottom)	Elevated 48dp, #2563EB	Submits roster to API & pops back
Take Group Photo	Session Screen (Actions)	Elevated with camera icon	Launches native camera for face capture

Upload Photo	Session Screen (Actions)	Outlined with gallery icon	Opens image gallery picker
Finish Session	Session Screen (Top Right)	Text button / Icon, #EF4444	Finalizes live class attendance session
Day Selector (Mon-Sun)	Timetable Screen (Top)	7 segmented pills, #2563EB	Filters timetable cards by weekday
Sign Out	Profile Screen (Bottom)	Elevated 48dp, #EF4444	Opens Confirm Sign Out Dialog

6. Designer Handoff & Implementation Recommendations

• **Minimum Touch Target Area:** Ensure all interactive touch targets (e.g. [P], [A], [OD] chips, icon buttons, day selector tabs) maintain at least 48x48dp clickable hit-test boundaries to prevent mis-clicks during fast-paced classroom roll calls.

• **High Contrast Standards:** The dark canvas (#0F172A) combined with pure white headings (#F8FAFC) delivers a contrast ratio of >13:1, meeting WCAG AAA requirements. Colored status chips must retain bold white labels for optimal readability in varying ambient room lighting.

• **Keyboard Auto-Dismiss:** On both the Login Screen and Attendance Entry Screen, tapping outside input text fields must dismiss the soft keyboard so the sticky submit bar remains accessible.

• **150ms State Transitions:** Interactive chips ([P], [A], [OD]) and navigation tabs utilize 150ms color transitions for immediate tactile feedback.

• **Dynamic Text Wrapping:** Long course names (e.g. 'Secure Database Management Systems') must wrap cleanly across multiple lines (softWrap: true) to avoid RenderFlex overflow errors.